

COSINE RULE

Higher | Mixed application extension

Additional problems combining trigonometry and geometry

Name

Date

2 questions

10 marks

2017-2025

Instructions

- Answer each question in the space provided.
- Show all your working and follow the rounding instructions.
- Diagrams are not accurately drawn unless stated otherwise.
- Use a calculator except for pages marked NON-CALCULATOR.

Using this booklet

Questions are arranged by exam series. Each page gives its original paper reference and question number. The question numbers within the extracted artwork are the original exam numbers.

These two additional problems can be solved using the cosine rule. Other valid methods also work. Use this extension after the main Higher booklet.

This is an independently compiled practice resource, not an official Pearson examination paper. The source collection covers 2017-2025. Questions are included only in the booklet that matches their main purpose.

Question index

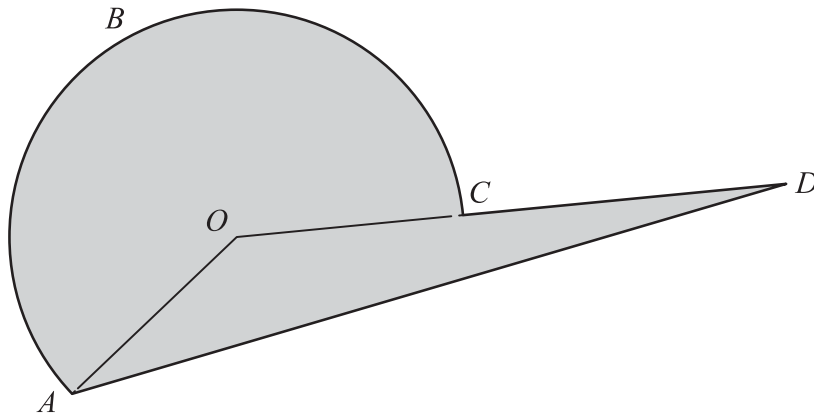
Higher | Mixed application extension | 2 questions | 10 marks

Practice	Exam series	Paper	Original question	Marks	Page
01	November 2018	1MA1/3H	Q16	5	3
02	November 2022	1MA1/3H	Q26	5	4

The original exam numbering is retained in each question.

Paper 1 questions are marked NON-CALCULATOR on their pages.

16 Here is a shaded shape $ABCD$.



The shape is made from a triangle and a sector of a circle, centre O and radius 6 cm.
 OCD is a straight line.

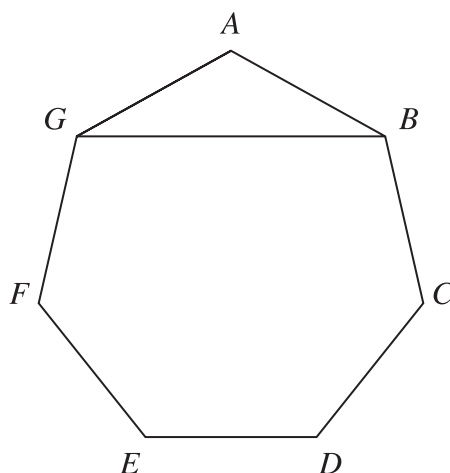
$AD = 14$ cm
 Angle $AOD = 140^\circ$
 Angle $OAD = 24^\circ$

Calculate the perimeter of the shape.
 Give your answer correct to 3 significant figures.

.....cm

(Total for Question 16 is 5 marks)

26 $ABCDEFG$ is a regular heptagon.



The area of triangle ABG is 30 cm^2

Calculate the length of GB .

Give your answer correct to 3 significant figures.

You must show all your working.

..... cm

(Total for Question 26 is 5 marks)
